THE DIAGONAL OF THE FINITE PART IS AN OPEN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Representability of diagonals of spaces. Spaces, section representable (cross-chapter, structural). The absolute diagonal factors through the base diagonal.
- Finiteness implies properness. Morphisms of Spaces, lemma finite-proper (cross-chapter, named). Makes the image E of the complement closed.

Source reference. Original equation 04PC: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1519–1519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NZ: focused-seeds/geometry/S9-smg-finite-diagonal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 025U: focused-seeds/geometry/S9-smg-finite-diagonal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The diagonal of $(X/Y)_{fin} \to Y$*

$$(X/Y)_{fin} \longrightarrow (X/Y)_{fin} \times_Y (X/Y)_{fin}$$

is representable (by schemes) and an open immersion and the "absolute" diagonal

$$(X/Y)_{fin} \longrightarrow (X/Y)_{fin} \times (X/Y)_{fin}$$

is representable (by schemes).

Proof. The second statement follows from the first as the absolute diagonal is the composition of the relative diagonal and a base change of the diagonal of Y (which is representable by schemes), see Spaces, Section 025U. To prove the first assertion we have to show the following: Given a scheme T and two pairs (a, Z_1) and (a, Z_2)

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over T with identical first component satisfying 04PC there is an open subscheme $V \subset T$ with the following property: For any morphism of schemes $h : T' \to T$ we have

$$h(T') \subset V \Leftrightarrow \Big(T' \times_T Z_1 = T' \times_T Z_2 \text{ as subspaces of } T' \times_Y X\Big)$$

Let us construct V. Note that $Z_1 \cap Z_2$ is open in Z_1 and in Z_2. Since $\mathrm{pr}_0|_{Z_i} : Z_i \to T$ is finite, hence proper (see Morphisms of Spaces, Lemma 04NZ) we see that

$$E = \mathrm{pr}_0|_{Z_1}(Z_1 \setminus Z_1 \cap Z_2)) \cup \mathrm{pr}_0|_{Z_2}(Z_2 \setminus Z_1 \cap Z_2))$$

is closed in T. Now it is clear that $V = T \setminus E$ works. $\square$